

GrounUp Enterprise Specification

Phase 23 - Software Development Foundation, Reference Architecture & Repository Bootstrap

Version 1.0.0 | Review-ready | Architecture PROPOSED - NOT APPROVED

1. Mission

Create the implementation repository foundation that translates GES requirements into governed software components, contracts, tests, environments and developer/AI execution rules.

2. Source Authority

Phase 23 is subordinate to the Phase 22 baseline, including 4,785 preserved source requirements and 24 developer/AI execution controls. It does not mark Phase 22 findings or final gates as approved.

3. Architecture Position

The reference architecture is modular, multi-tenant, contract-first, auditable and AI-provider-abstracted. Specific frameworks, cloud providers and database vendors remain proposed until architecture approval.

4. Repository Strategy

A governed monorepo blueprint versions applications, shared contracts, services, infrastructure, tests and architecture decisions together. Domain boundaries preserve extraction seams where independent scaling or isolation later justifies separate services.

5. SaaS/Tenancy

Tenant context is explicit across identity, authorization, transactional data, cache/queues, object storage, search, AI requests, events and audit evidence. White-label behavior is configuration-driven rather than copied business logic.

6. Construction Intelligence

Estimating, takeoff, production, cost, pricing, haul/disposal and regionalization are explicit domain/engine boundaries connected to governed master libraries and evidence-bearing plan intelligence.

7. AI Foundation

AI orchestration separates provider/model selection from business workflows and requires model/prompt/tool versioning, evaluation policy, source evidence, confidence/review handling and human approval boundaries.

8. Delivery Controls

CI/CD quality gates cover formatting/lint, unit/integration/contract/E2E tests, tenant isolation, security scanning, migrations, performance, AI evaluation, traceability and release evidence.

9. Definition of Done

A feature is complete only when its requirement references, code, tests, migrations, documentation, observability, security/tenant controls, traceability and acceptance evidence are complete.

Architecture Components

ID	Component	Layer	Purpose
P23-COMP-001	Web Application Shell	Presentation	Responsive SaaS web shell, navigation, tenant context, feature flags
P23-COMP-002	Mobile/Field Shell	Presentation	Field-optimized responsive/PWA boundary; native client may be added later
P23-COMP-003	API Gateway/BFF	Edge	Authenticated API entry, routing, rate limits, request correlation
P23-COMP-004	Identity Service	Platform	Authentication, SSO/MFA integration boundary, session/token policy
P23-COMP-005	Authorization Service	Platform	RBAC/ABAC/policy evaluation with tenant and project scopes
P23-COMP-006	Tenant Service	Platform	Tenant provisioning, isolation, entitlement and lifecycle
P23-COMP-007	Configuration Service	Platform	Tenant/system configuration, feature flags, branding
P23-COMP-008	Audit Service	Platform	Immutable audit/event evidence for governed actions
P23-COMP-009	Workflow Service	Platform	Approvals, states, assignments, escalations, evidence
P23-COMP-010	Notification Service	Platform	Email/SMS/push/in-app adapter boundary
P23-COMP-011	File/Object Service	Platform	Documents, drawings, photos, attachments, version metadata
P23-COMP-012	Search Service	Platform	Permission-aware search/index abstraction
P23-COMP-013	Master Library Service	Domain	Services/materials/resources/assemblies/rates/library governance
P23-COMP-014	Estimating Service	Domain	Estimate lifecycle, line items, assemblies, pricing orchestration
P23-COMP-015	Takeoff Service	Domain	Quantities, measurements, source evidence, takeoff state
P23-COMP-016	Production Engine	Domain	Crew/equipment/truck production calculation boundary
P23-COMP-017	Cost Engine	Domain	Labor/equipment/material/subcontract cost rollups
P23-COMP-018	Markup/Pricing Engine	Domain	Markup, O&P, contingency, price policy orchestration
P23-COMP-019	Haul/Disposal Engine	Domain	Trip-based haul, trucking, disposal and distance logic
P23-COMP-020	Regionalization Engine	Domain	Location factors, profiles, vendor/rate sources
P23-COMP-021	Project Service	Domain	Project/job lifecycle and shared project context
P23-COMP-022	CRM Service	Domain	Accounts, contacts, leads, opportunities and follow-up
P23-COMP-023	Workforce Service	Domain	People, crews, skills, labor assignments
P23-COMP-024	Fleet Service	Domain	Equipment, vehicles, utilization, maintenance, fuel hooks
P23-COMP-025	Procurement Service	Domain	Vendors, RFQs, POs, receipts, commitments
P23-COMP-026	Finance Adapter	Domain/Integration	Accounting/job-cost integration boundary
P23-COMP-027	Scheduling Service	Domain	Activities, dependencies, calendars, resource loading
P23-COMP-028	Commercial Service	Domain	Contracts, changes, claims, payment/commercial records
P23-COMP-029	Safety Service	Domain	HSE records, inspections, incidents, controls
P23-COMP-030	Document Control Service	Domain	Submittals/RFIs/drawings/specs/version control
P23-COMP-031	GIS/Reality Service	Domain	Spatial assets, surfaces, imagery/reality capture metadata
P23-COMP-032	Machine Data Adapter	Integration	Machine control/telematics integration boundary
P23-COMP-033	AI Orchestrator	AI	Model/provider abstraction, tool routing, policy, evidence
P23-COMP-034	Plan Intelligence Worker	AI	Plan/spec parsing, sheet/page evidence, confidence/review
P23-COMP-035	AI Evaluation Service	AI	Regression/evaluation datasets, scores, release gates
P23-COMP-036	Integration Hub	Integration	External connector/adapters, retries, idempotency

ID	Component	Layer	Purpose
P23-COMP-037	Event Bus Abstraction	Integration	Domain-event publication/consumption boundary
P23-COMP-038	Relational Data Store	Data	Transactional canonical records with tenant boundaries
P23-COMP-039	Cache/Queue Abstraction	Data	Ephemeral cache, jobs, locks, throttling
P23-COMP-040	Analytics Pipeline	Data	Operational-to-analytics governed data movement
P23-COMP-041	Observability Stack	Operations	Logs, metrics, traces, alerts, correlation
P23-COMP-042	Secrets/Key Boundary	Security	Secrets, keys, encryption integration abstraction
P23-COMP-043	CI Pipeline	DevOps	Build, lint, test, scan, package, provenance
P23-COMP-044	CD Pipeline	DevOps	Environment promotion, approvals, migrations, rollback
P23-COMP-045	IaC Repository	DevOps	Infrastructure definitions and environment policy
P23-COMP-046	Test Harness	Quality	Unit/integration/contract/E2E/performance/security test foundation
P23-COMP-047	Traceability Service/Index	Governance	Requirement-to-code-test-release evidence index
P23-COMP-048	Architecture Decision Registry	Governance	ADR lifecycle, supersession and owner approval

Architecture Decisions

ADR-0001 - Repository strategy [Proposed]: Use a governed monorepo blueprint so shared contracts, apps, services, tests and infrastructure version together. Rationale: Phase 22 requires traceability and controlled release.

ADR-0002 - Technology selection status [Proposed]: Do not treat a specific framework/cloud/database vendor as approved unless separately authorized. Starter code uses portable contracts and placeholders. Rationale: Source specification governs behavior more strongly than vendor choice.

ADR-0003 - Service boundaries [Proposed]: Use modular domain boundaries with a modular-monolith-first option and extraction seams for services requiring independent scale/isolation. Rationale: Reduces premature distributed complexity while preserving enterprise boundaries.

ADR-0004 - Tenant isolation [Required]: Tenant context must be explicit at identity, authorization, data, cache, object, search and audit boundaries. Rationale: Phase 22 execution controls require tenant isolation.

ADR-0005 - API/event contracts [Required]: Contract-first versioned APIs/events with compatibility tests and idempotency for integration operations. Rationale: Phase 22 API discipline.

ADR-0006 - AI provider abstraction [Required]: AI models/providers are replaceable behind governed orchestration, evaluation and evidence interfaces. Rationale: Avoid provider lock and preserve AI governance.

ADR-0007 - Audit/evidence [Required]: Governed business decisions and AI outputs preserve actor/source/version/time/evidence metadata. Rationale: Audit and traceability requirements.

ADR-0008 - Infrastructure portability [Proposed]: Separate logical platform contracts from cloud-specific implementation modules. Rationale: White-label/SaaS longevity and change control.